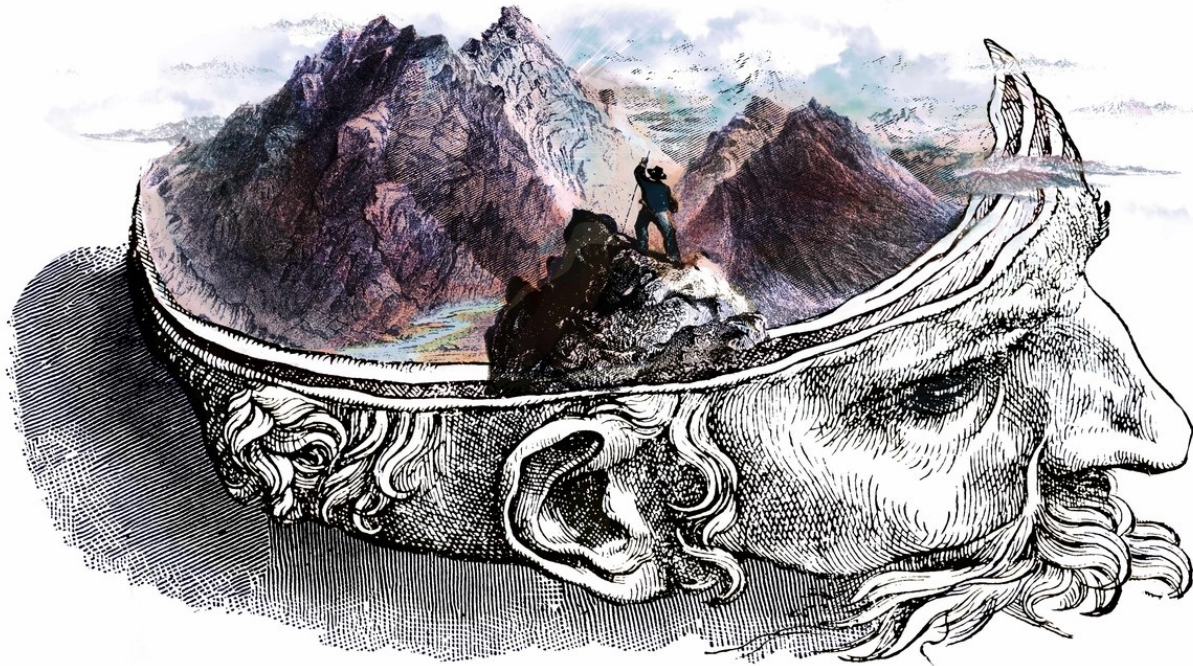


THE
NEW YORKER

THE MYSTERY OF S., THE MAN WITH AN IMPOSSIBLE MEMORY

By Reed Johnson August 12, 2017



The neuropsychologist Alexander Luria's case study of Solomon Shereshevsky helped spark a myth about a man who could not forget. But the truth is more complicated.

Illustration by Leigh Guldig

On an April afternoon in 1929, a timid-looking man with a broad face appeared at Moscow's Academy of Communist Education and asked to see a memory specialist. The man, who would become known in the psychological literature as S., had been sent by his boss, a section editor at a Moscow newspaper where S. was a reporter. That morning, the editor had noticed that S. did not take any notes when the daily assignments were made. When he confronted S. about this, S. explained that he didn't need to write anything down; he simply remembered. The editor picked up a newspaper and read at length from it, challenging S. to repeat everything back to him. When S. did so verbatim, the editor sent him to have his head examined.

The researcher who met with S. that day was twenty-seven-year-old Alexander Luria, whose fame as a founder of neuropsychology still lay before him. Luria began reeling off lists of random numbers and words and asking S. to repeat them, which

he did, in ever-lengthening series. Even more remarkably, when Luria retested S. more than fifteen years later, he found those numbers and words still preserved in S.'s memory. "I simply had to admit that the capacity of his memory had *no distinct limits*," Luria writes in his famous case study of S., "The Mind of a Mnemonist," published in 1968 in both Russian and English. In the book, Luria describes how S., desperate to purge his mind of unwanted recollections, turned to writing down everything he wanted to forget on slips of paper, in the hope that he might somehow offload these memories. When this failed, he lit the slips of paper on fire and watched them burn to ash, also to no avail.

Luria's monograph became a psychology classic both in Russia and abroad, and it had considerable influence over the nascent field of memory studies. S.'s case became a parable about the pitfalls of flawless recall. Luria catalogues various difficulties that S. experienced navigating everyday life, linking them to profound deficits he identified in S.'s ability to conceive the world in abstract terms. These cognitive deficiencies, Luria suggests, were related to S.'s extraordinary episodic memory—the memory we have for personal experiences, as opposed to semantic memory (which tells us, for instance, that the dromedary has only one hump). Deriving meaning from the world requires us to relinquish some of its texture. S.'s case, as many readers have noted, resembles the Jorge Luis Borges story "Funes the Memorious," a fictional work about a man plagued by the persistence of his memory. "To think is to forget a difference, to generalize, to abstract," Borges writes. "In the overly replete world of Funes there were nothing but details, almost contiguous details." Similarly, Luria writes that for S., almost every word, every thought, was freighted with excessive detail. When he heard "restaurant," for example, he would picture an entrance, customers, a Romanian orchestra tuning up to play for them, and so on. Like Funes, S. had a sort of private language to catalogue the richness of his mental associations. The word for "roach" in Yiddish could also mean, in his mind, a dent in a metal chamber pot, a crust of black bread, and the light cast by a lamp that fails to push back all the darkness in a room.

With its richly descriptive prose, Luria's case study reads as a literary work as much as a psychological monograph. Luria wanted to discover not so much what S. was like but, to repurpose the philosopher Thomas Nagel's famous formulation about a bat, to discover what it was like for S. to be S. He quotes long passages from their interviews and correspondence, and the two voices—Luria, measured and thoughtful; S., exuberant, full of almost hallucinatory imagery—intermingle on the

page, offering glimpses of a strange inner landscape. Luria is more reticent when it comes to the outer world of the man he studied for nearly three decades. He offers only the barest of biography, and never identifies S. by his real name in the book—even though S. had been dead for a decade when Luria published his study and his memory feats were already well known in the U.S.S.R. Exactly how S. died, or what he was doing in his later years, Luria doesn't say. Some sources have him spending his sunset years as a Moscow cabbie, ferrying passengers around without need for a map, while others assert that he went insane and ended up in an asylum, unable to distinguish the present from the ever-living past of his memory. Neither, it turns out, is true.

For years now, since first reading Luria's book as an undergraduate studying Russian, then after encountering it again as a research assistant in a memory lab, I've searched, on odd weekends and nights, for what information I could find about S., whose real name was Solomon Shereshevsky. Eventually, I tracked down a relative. Then, more recently, I got hold of a small, blue school notebook, preserved by Luria's grandniece in the psychologist's archives. It contains Shereshevsky's own handwritten autobiographical account of how he became a mnemonist. Written not long before his death and left incomplete, it opens with his impressions of that first meeting with Luria twenty-eight years earlier. It even provides the exact list of things Luria gave him to memorize that day.

My search for Solomon Shereshevsky revealed a person who fit uneasily in the story of the Man Who Could Not Forget, as he has so often been portrayed. He did not, in fact, have perfect recall. His past was not a land he could wander through at will. For him, remembering took conscious effort and a certain creative genius. He was not a photographer, I've come to think, so much as an artist—a person who painted not from memory but *with* memory, combining and recombining his colors to make worlds only he could see. His extraordinary case also reveals something of how our ordinary minds remember, and how often they do not.

S hereshevsky's own account of his life as a mnemonist diverges from Luria's on its very first page. He dates their meeting to April 13, 1929, while Luria has it occurring a few years before, and Shereshevsky gives his age at the time as thirty-seven, while Luria asserts that his subject was still in his twenties. According to Shereshevsky, he returned to the newspaper that day and told his editor that his memory had been tested and was found to exceed the bounds of what was believed to be physically possible. Hearing this, the editor convinced him to give up writing

—at that time, Shereshevsky's specialty was brief satirical pieces that, in the early years of Stalin's rule, had fallen from favor—and instead devote himself to performing full time as a professional mnemonist. In short order, he hired a circus trainer as his manager and travelling assistant and was coached by a carnival juggler on how to entertain. Then he set off for the provinces.

Mikhail Reynberg attended one of Shereshevsky's performances in a small town outside Moscow. For Reynberg, S. was simply Uncle Solomon. I tracked him down through a contact in Moscow and went to see him a few years ago on a hot summer afternoon in Brooklyn, where he now lives. His apartment was a rambling series of neatly kept rooms that had an unmistakably Russian feel to them, from the beaded hallway curtain to the feast of delicious *zakuski* that had been laid out for my arrival. Reynberg is stocky, with a head of neatly combed ivory hair, and we sat for hours in the kitchen, talking about his uncle. In that town outside Moscow, he said, the farmers who were supposed to meet him and his uncle at the train station never showed up, so they hired horse-drawn sledges to take them through the snow to find the venue on their own. When they arrived, the organizers had set up the chalkboard for Shereshevsky's performance, but then had broken into the refreshments meant for the reception and had gotten drunk. Uninterested in testing the limits of the mnemonist's mind, the villagers instead convinced their visitors to join them around the table and continue drinking. The actual performance never happened, but they paid him nonetheless—in potatoes, Reynberg recalled, for which his uncle was grateful.

VIDEO FROM THE NEW YORKER

The Absurdity of the Trump-Putin Summit in Helsinki

Like food, housing was in short supply in those years. In Moscow, Shereshevsky lived with his wife and son in a damp room in the basement of a janitorial outbuilding tucked away in a courtyard. It was a trying situation for them all, but perhaps especially for Shereshevsky's wife, Aida. A graduate of the famous Smolny Institute for Noble Maidens, Aida was a talented musician who kept her own piano in their cramped quarters. During spells of fine weather, husband and wife wheeled the instrument out into the courtyard to let it dry out. There was something striking about this incongruous image: the two of them trundling the heavy piano to a sunny spot in the courtyard, each bump calling forth dim polychromatic echoes from inside its wooden body.

Reynberg insisted that Shereshevsky's lot in life was not merely bad luck but the result of an orchestrated campaign of intimidation. According to Reynberg, Shereshevsky was pressured to put his talents to work for the secret police, but he declined. His problems deepened after the Second World War, Reynberg said, during the so-called anti-cosmopolitanism campaign, a purge directed primarily at Jews. (Luria, by then a well-known psychologist, was hounded from his job at the Institute of Neurosurgery, and, according to his daughter's family memoirs, kept a suitcase packed and ready for his potential arrest.) Shereshevsky found himself increasingly shunned, his shows cancelled. Reynberg told me that the secret police planted provocateurs in Shereshevsky's audiences, some of whom heckled and whistled at him in order to distract him from his routine. After a disastrous

performance that left audience members clamoring for a refund, his career was essentially finished. That career neatly overlapped with Stalin's rule, spanning several waves of mass arrests. Hundreds of thousands, even millions, vanished into the maw of the camps; disgraced public figures were literally airbrushed from the record; the dead and disappeared were purged not just from life but from the nation's collective memory. Shereshevsky had made a living off his memory in a land ruled by amnesia.

Something else I learned that afternoon threatened to change my entire sense of who Shereshevsky was: His uncle, Reynberg said, could be forgetful. If he didn't consciously try to commit something to memory, he didn't always recall it later. I had imagined, based on Luria's case study and the mythology that had grown up around it, a Soviet Funes, with flawless and involuntary recollection of his past. Reynberg told me that his uncle trained hours a day for his evening performances. Was he a mere showman after all?

As described by Luria, some of Shereshevsky's mental operations bear a strong resemblance to the sort of garden-variety mnemonic tricks that have been known for many centuries—for example, the “memory palace,” or “method of loci,” in which an imagined physical space is used to organize information in its proper sequence. In Shereshevsky's version of this device, he would imagine Gorky Street, Moscow's main thoroughfare, or a village street from his childhood, mentally distributing what he wanted to remember along its length, often creating an impromptu story out of the sequence, then strolling back through later to recollect (re-collect) these items in his mind. It's a technique that nearly anyone can learn; in his book “Moonwalking with Einstein,” the writer Joshua Foer describes using this technique to win at the U.S.A. Memory Championship. Luria doesn't deny Shereshevsky's use of mnemonic devices, but he maintains that these came later, and that they merely complemented Shereshevsky's immense natural abilities.

Luria also notes that Shereshevsky had an extraordinarily strong case of synesthesia, the heritable condition in which the senses become intermingled in the mind, and the psychologist recognized that this had something to do with Shereshevsky's powers of recall. (Vladimir Nabokov wrote about both his “colored hearing” and exceptional recall in his memoir, “Speak, Memory,” first published in 1951.) When Luria rang a small bell, for instance, the sound would evoke in his subject's mind “a small round object . . . something rough like a rope . . . the taste of salt water . . . and something white.” Shereshevsky thought of numbers in the same colors and fonts that he first saw them in as a child; in his unpublished notebook, he writes that “all

the numbers had names, first and last, and nicknames, which changed depending on my age and mood.” The number one “is a slender man with ramrod posture and a long face; ‘two’ is a plump lady with a complicated hairdo atop her head, clad in a velvet or silk dress with a train that trails behind her.” Luria speculates that Shereshevsky used his web of multimodal associations to cross-check his memory.

There were serious drawbacks in having so many channels open to the world. Shereshevsky avoided such things as reading the newspaper over breakfast because the flavors evoked by the printed words clashed with the taste of his meal. And as he stood in front of his crowds, memorizing long streams of nonsensical information—cantos from Dante’s *Inferno* in the original Italian, for instance, a language he did not speak—he seemed to find that his natural abilities could hinder as much as help him. Much like a professional athlete, Shereshevsky had to change and relearn formerly unconscious strategies in order to grapple with ever-larger challenges. The strength and durability of his memories seemed to be tied up in his ability to create elaborate multisensory mental representations and insert them in imagined story scenes or places; the more vivid this imagery and story, the more deeply rooted it would become in his memory. But with only a few seconds for each item on his list, he could no longer allow the same flow of spontaneously evoked imagery as before. Instead, he had to control, to standardize, to essentialize. Gone were the Romanian orchestras tuning up their instruments in his mind; in their place, the word “restaurant” became a flash of white tablecloth.

What stands out most upon rereading Luria’s book is not Shereshevsky’s ability to rattle off series of numbers and words but the intricate sense impressions that, for him, bloomed behind each syllable in those series: a dusty village street in Rezhitsa, his old landlady yelling out a window in Yiddish to the rag-and-bones man, three jackdaws roosting in an old tree. Luria’s famous case study of extraordinary memory turns out to be less about perfect recall and more about something at once more fundamental and more strange: our ability to conjure such sensory details even without the direct input of our senses, to swim against the usual currents of perceiving minds. It’s this same ability that allows us to daydream, or to do thought experiments in physics, or to read words on a page and hear the dim inner echo a piano makes when rolling across the cobbles of a Moscow courtyard. But what do imagination and make-believe have to do with memory—a mental faculty we value precisely for its supposed veracity?

his question was one that I recently posed to Daniel Schacter, a professor of psychology at Harvard and the author of “The Seven Sins of Memory: How the Mind Forgets and Remembers,” among other books. We met this past spring at the University of Virginia, at a conference that paired literary scholars with neuroscientists to discuss the workings of memory. Schacter’s interest in the connection between memory and imagination stretches back to the nineteen-eighties, when he and his mentor, Endel Tulving, interviewed a profoundly amnesiac patient known by the initials K. C. A victim of a motorcycle accident, K. C. had become incapable of forming episodic memories. He couldn’t say what he was doing a day or even an hour prior. He was also, somewhat unexpectedly, unable to speculate in any detail about what he would be doing the *following* day. He couldn’t call up any detailed scenes in his mind’s eye, whether these scenes lay in the actual past or in some imagined future.

There was a connection, his case seemed to imply, between memory and imagination. But how could one study that? At the time, Schacter and his colleagues didn’t have many tools. But the next two decades saw the advent of MRI technology that allowed researchers to see mental processes as they were unfolding in the brain. Neuroimaging showed that patterns of brain activation for episodic memory and imagining the future were virtually indistinguishable. The discovery fuelled a paradigm shift in memory research; one review of the field found that over a recent five-year period, the number of articles published on memory and imagination had increased tenfold. The experimental evidence suggests to Schacter that our imagination draws heavily on memory, recombining bits and pieces of actual experience to model hypothetical and counterfactual scenarios. This seems intuitive. But he goes further, arguing that our all-too-fallible recollections of the past are in fact adaptive, providing the flexibility that allows us to reconfigure memory to imagine our possible futures.

And the relationship might run the other way, too: as Schacter noted when we spoke, the directionality of the interaction between memory and imagination has not yet been established. “This is an area that is just still in its infancy,” he said. “The past decade has just gotten the engine started, and it’s going to take off from here.” Certainly, our memories are more beholden to imagination than we’d like to think—the pioneering memory researcher Elizabeth Loftus has spent a career showing how easily one can implant false memories in subjects through guided mental imagery of made-up events, for example. The creation of false memories is perhaps not entirely

unlike Shereshevsky's visualization of made-up scenes in various physical locations, his personal variation on the "memory palace" technique. The secret to the success of that old mnemonic device may, in fact, have something to do with our underlying brain structures. Researchers have discovered that the hippocampus is not only a seat of memory but also the site in the brain where we create mental maps of our world. In "The Mind of a Mnemonist," Luria notes that Shereshevsky's father, a bookseller, knew the exact location of every one of the books on his store's shelves; one wonders whether a talent for spatial memory was passed on from father to son. One theory of imagination and memory proposed by neuroscientists Demis Hassabis and Eleanor Maguire suggests that the hippocampus plays a central role in constructing mental scenes, and that it is precisely the ability to choreograph these scenes in our minds that allows us both to reexperience the past and to imagine the future.

When you reinterpret Shereshevsky's story as a case of an extraordinary imagination, other aspects of his mind come into sharper focus. Some of these are oddities that are left out of most accounts of his life, perhaps because they fit poorly into the Funes-style narrative. Luria relates that Shereshevsky was capable of sitting in a chair and consciously modifying his heart rate from sixty-four beats per minute to a hundred by picturing himself either lying in bed or racing after a train just leaving the station, respectively. According to Luria's experiments, Shereshevsky could alter the skin temperature of his hands by several degrees by visualizing himself touching a hot stove or a block of ice. Imagining a loud noise caused an involuntary protective reflex in his eardrums, as though the sound had actually occurred.

Given the verisimilitude of Shereshevsky's inner world, it's perhaps not surprising that he was apt to confuse his imagination with reality. He recounts to Luria how, as a child, he might lie in bed past the time when he needed to get up for school, having imagined that the clock had stopped. He could also, he tells Luria, imagine himself splitting into a doubled self, one of whom had to go to school while the other stayed home. Luria describes this vivid daydreaming as fatal to Shereshevsky's ability to function in the adult world. Like the daydreaming narrator of Dostoyevsky's "White Nights," Shereshevsky constructed a more riveting reality in his mind; when reality failed to measure up, he retreated further into fantasies. "I read a great deal and always identified myself with one of the heroes," he told Luria, in 1937. "For I saw them, you know." He continued, "Even at eighteen I couldn't understand how one friend of mine was content to become an accountant, another a travelling salesman." He believed he was "destined for something greater," though he

couldn't say what this might be—his dreams for the future changed frequently, though he was unable to do the work required to realize them. “That’s the way I’ve always been,” he told Luria.

Shereshevsky strongly disputed Luria’s implication that he suffered from a mental pathology. In his notebook, he writes that he consented to a further series of experiments at Moscow’s Hospital for Diseases of the Nervous System, in the hope that they would provide him with a clean bill of health. It’s not clear if one was ever given. There’s a palpable sense of frustration in Shereshevsky’s recollections, an undercurrent hinting at fraught relations with his experimenters, who, Shereshevsky writes, kept insisting that he must be concealing from them some secret technique, some “hidden combination” that would explain his mental abilities.

Instead of burning memories on scraps of paper, Shereshevsky found a different kind of erasure in his final years, according to his nephew: he turned to drinking. He did not, as it has been claimed, end up in an insane asylum, though his drinking may have been an expression of what Soviet citizens called “internal emigration.” As Luria writes, “One would be hard put to say which was more real for him: the world of imagination in which he lived, or the world of reality in which he was but a temporary guest.” Shereshevsky died in 1958 from complications related to his alcoholism. The last dated entry in his notebook is from December 11, 1957, but in it he writes only of the past, remembering experiments and performances from his earlier days as a professional mnemonist. After that, it breaks off into blank pages, as though inviting us to imagine other possible endings.

Video

Diana Vishneva's Last Days with American Ballet Theatre

Vishneva, a principal dancer with American Ballet Theatre, rehearses for the Russian ballet "Onegin," which will be her final performance with the company.

CONDÉ NAST

© 2018 Condé Nast. All rights reserved. Use of and/or registration on any portion of this site constitutes acceptance of our [User Agreement \(updated 5/25/18\)](#) and [Privacy Policy and Cookie Statement \(updated 5/25/18\)](#). Your [California Privacy Rights](#). The material on this site may not be reproduced, distributed, transmitted, cached or otherwise used, except with the prior written permission of Condé Nast. *The New Yorker* may earn a portion of sales from products and services that are purchased through links on our site as part of our affiliate partnerships with retailers. [Ad Choices](#)